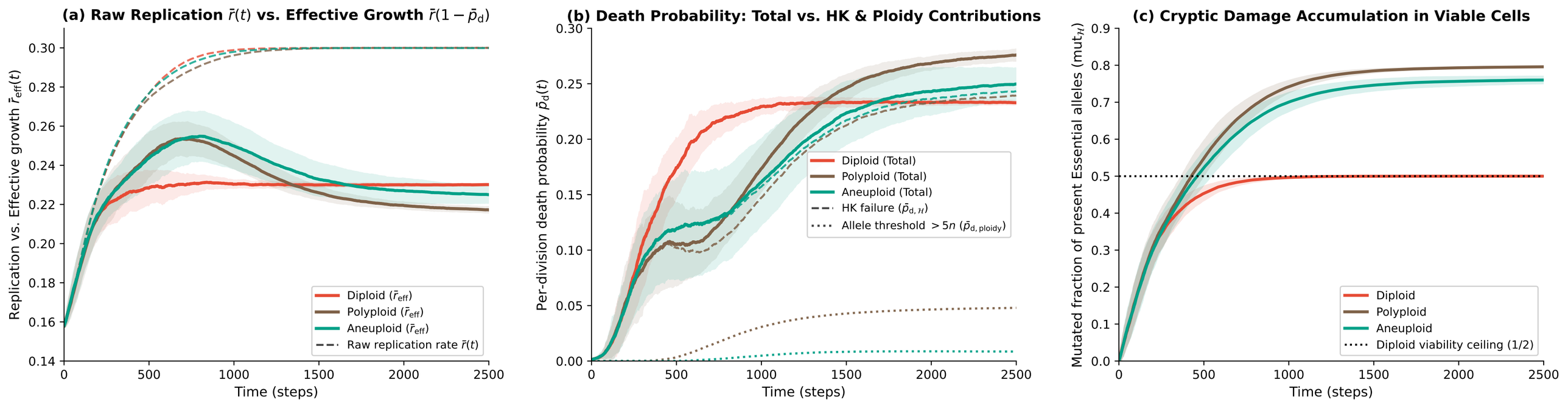

(a) Raw Replication $\bar{r}(t)$ vs. Effective Growth $\bar{r}(1-\bar{p}_d)$
Replication vs. Effective growth $\bar{r}_{eff}(t)$
Time (steps)
Diploid ($\bar{r}_{eff}$)
Polyploid ($\bar{r}_{eff}$)
Aneuploid ($\bar{r}_{eff}$)
Raw replication rate $\bar{r}(t)$
(b) Death Probability: Total vs. HK & Ploidy Contributions
Per-division death probability $\bar{p}_d(t)$
Time (steps)
Diploid (Total)
Polyploid (Total)
Aneuploid (Total)
HK failure ($\bar{p}_{d,\,H}$)
Allele threshold $> 5n$ ($\bar{p}_{d,\,ploidy}$)
(c) Cryptic Damage Accumulation in Viable Cells
Mutated fraction of present Essential alleles (mut_H)
Time (steps)
Diploid
Polyploid
Aneuploid
Diploid viability ceiling (1/2)